NSO8062 // NSO7007 Geofüüsikaline andmetöötlus

Loeng 2: Erinevad andmete tüübid ja lihtsamad tegevused nendega

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Ajakava ja teemad

Nädal	Teema	Õppejõud
1	Sissejuhatus ainesse	S-T. Luik
2	Erinevad andmete tüübid ja lihtsamad tegevused nendega.	S-T. Luik
3	NetCDF failide loomine, muutmine jne.	S-T. Luik
4	Keri andmed. Profiilide visualiseerimine, andmete eemaldamine, silumine, gradientide leidmine.	S-T. Luik
5	Andmete 1D interpoleerimine ja korrelatsioonid, statistikud jne.	S-T. Luik
6	Tuuleandmed ja nende visualiseerimine	S-T. Luik
7	2D andmed, interpoleerimine, visualiseerimine (nt kaart)	
8	TBD	
9	TBD	
10	TBD	
11	TBD	
12	TBD	
13	TBD	
14	TBD	
15	Konsultatsioon	
16	**Eksam**	
17		
18		
19	10/06/2026 kevadsemestri eksamite sooritamise lõpptähtaeg	
20	15/06/2026 kevadsemestri lõpp	

Lecture 2. Exercise 1.

- Write a code that creates two 5x1 numeric vectors.
- Add, subtract, multiply, and divide these vectors.

Start your code with:

```
clearvars % Clears workspace
close all % Closes figures etc.
format long % Longer number format for more precision
clc % Clears the command window
```

Save workspace

```
save('results_lec2_ex1.mat')
```

save code: Editor > Save as > code_lec2_ex1.m

Lecture 2. Exercise 2.

- Write a code that generates a random 20x30 matrix.
- Get statistics (min, max, mean, median, sum) of said matrix.
- Get statistics per row and column and per the entire matrix.
- Create a three level struct to hold your results.

  e.g.: a_stat.a_min.row

Save your code and only the struct and matrix vars.

rand / min / max / mean

Lecture 2. Exercise 3.

- Load your matrix (from Lecture 2, exercise 2).
- Write this matrix to a .txt file.
- Write this matrix to an .xlsx file.
- Clear your workspace and load the .txt and .xlsx files as separate variables.

Save your code.

writematrix / readmatrix / sheet / range / xlsfinfo

Lecture 2. Exercise 4.

Data type conversions etc. Create variables and convert them.

1) character to double and *vice versa*
2) string to double and *vice versa*
3) character to string to double
4) numeric vector to logical
5) character, string, and double arrays to cell arrays
6) datetime array from year, month, day, hour, minute, second and *vice versa*
7) arrays to table and arrays from table
8) matrix to cell and *vice versa*

Add descriptions to your code and save the code.

Lecture 2. Exercise 5.

Sorting arrays

- Create two numeric vectors. Randomize their order.

```
a = (1:0.4:30)';
a_rand = a(randperm(length(a))); % Vector in a random order
```

- Sort from smaller to bigger (ascend) and from bigger to smaller (descend).
- Create a table, use sortrows, specify columns used for sorting.
- Create a matrix, use sortrows, specify columns used for sorting.

Replacing values

- Replace some values in your array, matrix, table by indexing.
- Replace some values based on set conditions (==, ~=, <, >, >=, <=, contains, find …).

Add descriptions to your code and save the code.

Lecture 2. Exercise 6.

Loading tables, getting info and renaming variables

Info about the table before loading it:

```
opts = detectImportOptions('data_for_lec2_ex6.txt');
```

Load data:

```
T = readtable('data_for_lec2_ex6.txt');
```

Load only specific variables:

```
opts.SelectedVariableNames = ["Var1","Var2"];
T2 = readtable('data_for_lec2_ex6.txt',opts);
```

Summary of table – summary(T)

Table properties – T.Properties / T.Properties.VariableNames

Rename table vars – T_newname = renamevars(T1,vars,newNames)

height / width / head / tail

Add descriptions to your code and save the code.

Lecture 2. Exercise 7.

Joining tables (data_for_lec2_ex6.txt)

- Load data with readtable, get only coordinates and depth (table1).
- Load data again, get only depth, temperature, and salinity (table2).
  Sort table based on salinity data.
- Join tables
  join(table1,table2)
  join(table2,table1)
- Create a subset of both tables and:
  outerjoin(table1,table2)
  outerjoin(table1,table2,'MergeKeys',true)
  innerjoin(table1,table2)

Add descriptions to your code and save the code.

Join Type	Function	Description	Rows Included in Result	Missing Values Handling
Inner Join	innerjoin(T1, T2)	Returns only the rows with matching keys in both tables.	Only matching rows from both T1 and T2.	No missing values; non-matching rows are excluded.
Left Join	join(T1, T2)	Returns all rows from the left table (T1) and the matching rows from the right table (T2).	All rows from T1 and matching rows from T2.	NaN for non-matching rows in T2.
Right Join	join(T2, T1)	Returns all rows from the right table (T2) and the matching rows from the left table (T1).	All rows from T2 and matching rows from T1.	NaN for non-matching rows in T1.
Outer Join	outerjoin(T1, T2)	Returns all rows from both tables, filling in missing values where there are no matches.	All rows from T1 and T2.	NaN for non-matching rows in either table.
Full Outer Join	outerjoin(T1, T2, 'MergeKeys', true)	Similar to outer join, but merges keys from both tables into a single key column.	All rows from both tables, with merged keys.	NaN for non-matching rows in either table.